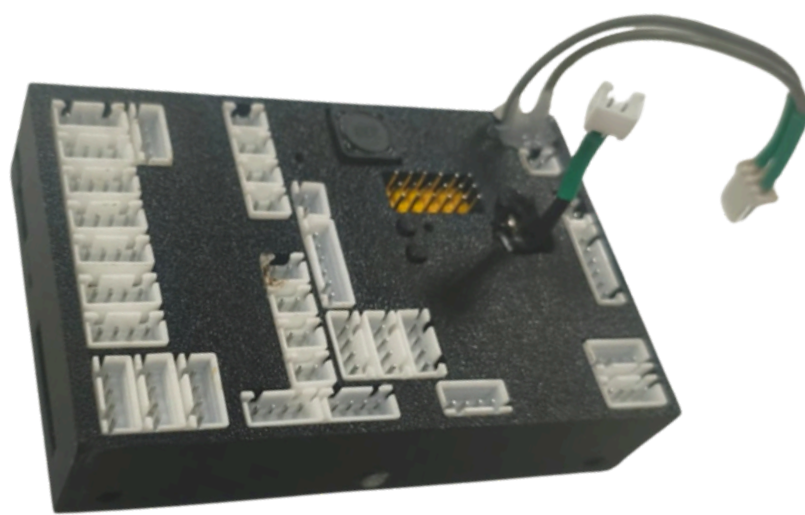


HARDWARE

The Ligeirinho was built entirely from scratch for RoboCup 2026, featuring an innovative double-front architecture. Instead of adapting commercial kits, the chassis is entirely 3D printed. We designed the structural core to lower the center of gravity and integrated hand-curing silicone tires on repurposed vacuum cleaner wheels, ensuring zero slippage on 25-degree ramps. To overcome ambient light restrictions during line following, we developed a fully enclosed, light-sealed dome for the lower camera.

CONTROLLERS

MAIN BOARD



Powered by a Banana Pi M4 Zero, this central unit handles all power and processing requirements. It is responsible for processing images from the dual-camera system, running the PID control loop, and executing AI models in real time.

SENSORS

REED SWITCHES



Quant.: 2

Magnetic sensors in the bumper for collision detection.

DISTANCE SENSOR



Quant.: 6

In the front, on the sides, and three in the robot's bucket.

GYROSCOPE



Quant.: 1

In the middle of the robot, it has a microcontroller to process the x, y, and z tilt planes.

CAMERA



Quant.: 2

Bottom camera for line PID (inside the sealing dome); Front camera for AI-powered victim and OpenCV-powered detections.

ACTUATORS



MG90S MICRO SERVO MOTOR

Quant.: 4

They control the victim-grabbing claw and the automated internal ramp of the split bucket. Have a modified extra plate for reading.



Traction Motors

Quant.: 4

They provide high torque for climbing ramps and overcoming obstacles.

TEAM



EDGAR ALVES ARRIGONI

Hardware Engineering and Manufacturing (3D modeling, mechanical structure and manufacturing of customized tires).



GABRIEL DOMINICINI DA FONSECA

Software Engineering and Control (Computer vision algorithms, PID tuning and AI model training).



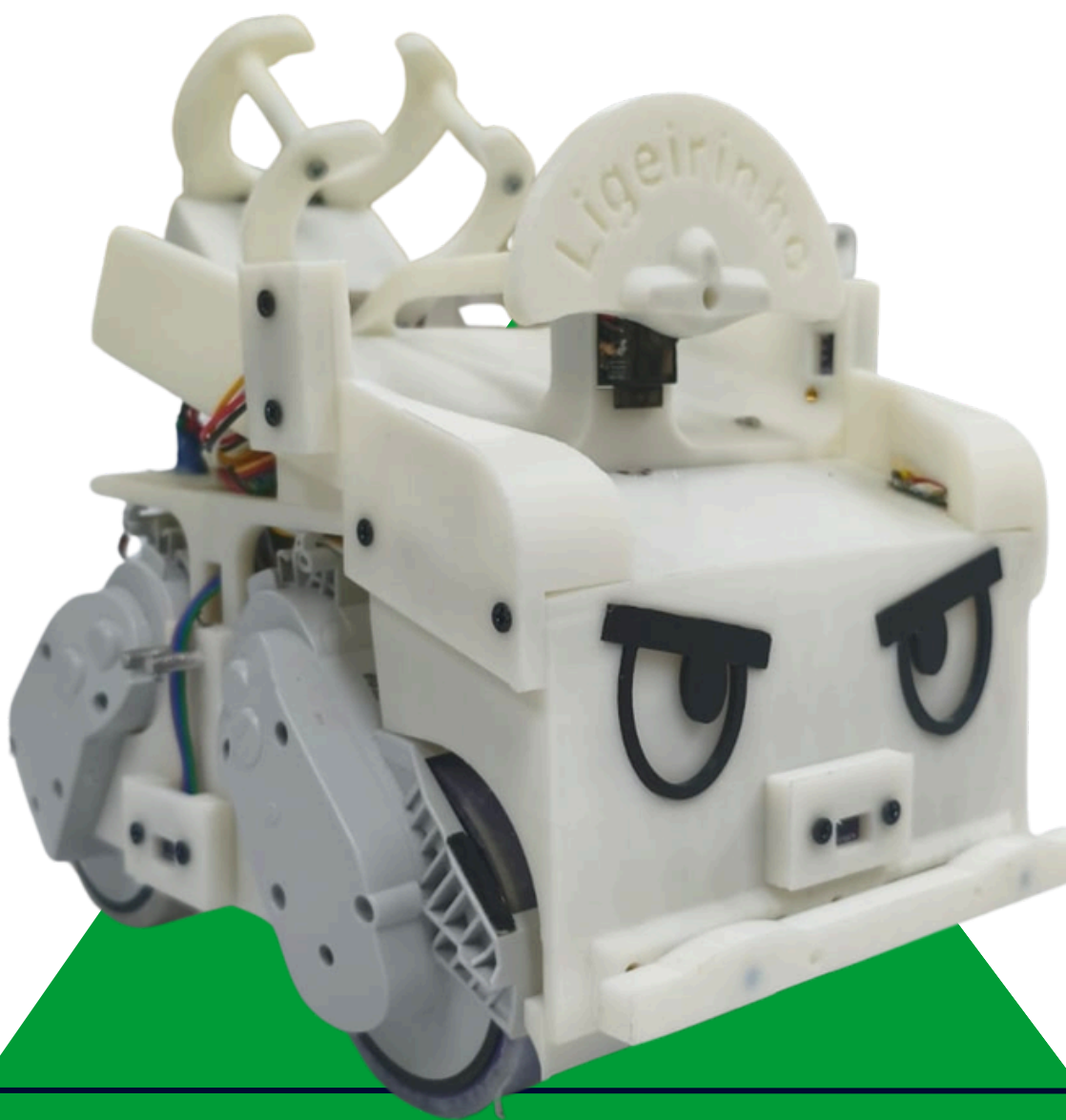
BERNARDO SCHMIEDEL RAYMUNDO

Full-Stack Mechatronics Leader (Custom electronics/PCB fabrication, hardware-software integration, and troubleshooting).



JULIO CESAR GOLDNER VENDRAMINI

Team Mentor.



LIGEIRINHO

OUR HISTORY

We are the oldest Team of Brasil on active and over time we have achieved:

- 2022 – 1st place in the State Championship (ES) and participation in the National Championship.
- 2023 – 1st place in the State Championship (ES) and 1st place in the National Championship (BR).
- 2024 – 5th place in the World Championship (RoboCup).
- 2025 – 1st place in the State Championship (ES), 2nd place in the National Championship (BR) and 1st place in the National Superteams Championship (BR).



1st PLACE OBR
STATE (ES) 2022



1st PLACE OBR
STATE (ES) 2023



1st PLACE OBR
NATIONAL (BR) 2023



5th PLACE
ROBOCUP RESCUE
LINE WORLD 2024



1st PLACE OBR
STATE (ES) 2025



2nd PLACE OBR and
1st PLACE
SUPORTEAMS
NATIONAL (BR) 2025

SOFTWARE

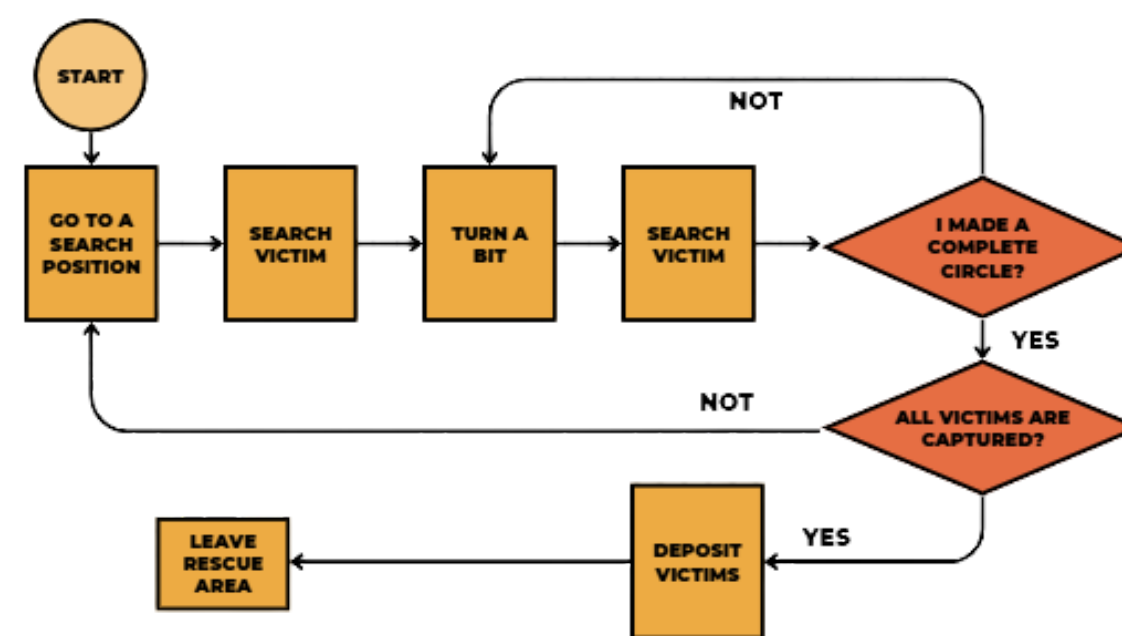
Our software operates based on a deterministic Finite State Machine (FSM).

For line tracking, we developed a high-speed PID control loop based on a virtual pixel array extracted from the bottom camera, allowing for precise 90-degree turns without traditional infrared sensors.

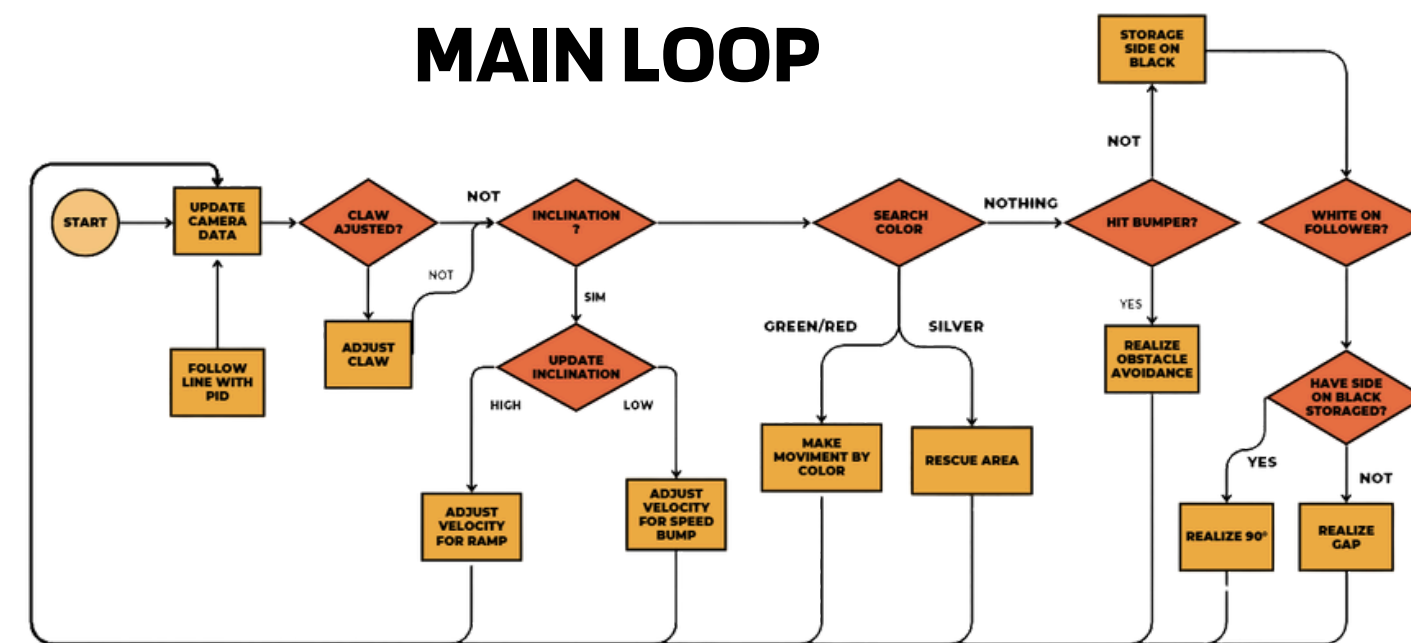
For the Rescue Area, we run a lightweight Convolutional Neural Network (CNN). This Artificial Intelligence model runs locally at a stable 3 FPS, instantly detecting and classifying victims (alive/dead) based on color to trigger our automated triage mechanism. For the triangles we use only OpenCV and algorithms, not AI.

ALGORITHM

RESCUE AREA



MAIN LOOP



HARDWARE INNOVATIONS



LIGHT DOME

A custom-printed 3D-printed fairing for the lower camera that completely blocks out external light, correcting reflection problems.



SILICONE TIRES

Hand-cured silicone tires mounted on vacuum cleaner wheels for maximum traction.

LIGHT DIFFUSER



The Light Dome features a built-in diffuser for the internal LEDs, ensuring perfectly uniform illumination and eliminating glare for the bottom camera's computer vision.

MOUNTING SIDE



Features a modular side bracket to quickly attach a screen and keyboard, allowing for on-the-fly debugging and code adjustments.